

Question:

Calculate the quantity of heat required to (a) convert a liter of H₂O at 30°C to a liter of water at 60°C, and (b) heat a 1 kg block of aluminum from 30° to 60°C. Assume the specific heat of water and aluminum is, re-spectively, 1 cal/g°C and .215 cal/g°C.

Answer Choices:

- A. 28 Kcal, 6.3 Kcal
- B. 40 Kcal, 5 Kcal
- C. 30 Kcal, 8.58 Kcal
- D. 25 Kcal, 8 Kcal
- E. 29 Kcal, 7.77 Kcal
- F. 34 Kcal, 6.9 Kcal
- G. 30 Kcal, 6.45 Kcal
- H. 32 Kcal, 7 Kcal
- I. 35 Kcal, 7.5 Kcal
- J. 31 Kcal, 6.15 Kcal

Reasoning + Answer:

<think>
water heat capacity 1 cal/g C
weight of 1 liter 8 . 3 4 5 5 5 gram
heat = mass Ã specific heat Ã â C
8 . 3 4 5 5 5 g â 1 cal C
8 3 4 . 5 5 5 â 1 0 0 0 cal K
for heat from 3 0 Â° C to 6 0 Â° C 9 0 C
9 0 C â 8 3 4 . 5 5 5 cal K â 1 0 0 0 cal K - 1
7 4 , 1 1 0 cal
energy needed first 7 4 , 1 1 0 cal
for second step ,
specific heat of Al is . 2 1 5 cal/g C
heat of Al , 3 0 Â° C to 6 0 Â° C
3 0 Â° C â 6 0 Â° C 3 0 C 1 kg = 1 0 0 0 g
Al heat is 1 0 0 0 g Ã . 2 1 5 cal/g C Ã 3 0 C = 6 4 5 cal
heat required 7 4 , 7 5 5 cal
answer B , 7 4 . 8 Kcal
</think>
< answer >
7 4 . 8 Kcal
</answer >